

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
1.579	80.0	227.0	230.0	4503.563	19.581

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[11,3]	[61,43]	109.669	BRANCH	BRANCH
2	[15,0]	[11,3]	3.828	TIP	BRANCH
3	[11,3]	[22,5]	10.414	BRANCH	BRANCH
4	[23,0]	[22,5]	4.414	TIP	BRANCH
5	[42,24]	[22,5]	28.627	TIP	BRANCH
6	[423,0]	[422,6]	5.414	TIP	BRANCH
7	[426,6]	[422,6]	3.0	TIP	BRANCH
8	[422,6]	[376,9]	55.627	BRANCH	BRANCH
9	[369,9]	[376,9]	6.0	TIP	BRANCH
10	[379,6]	[376,9]	2.828	TIP	BRANCH
11	[252,7]	[250,14]	6.828	TIP	BRANCH
12	[300,5]	[250,14]	52.728	TIP	BRANCH
13	[250,14]	[239,18]	11.657	BRANCH	BRANCH
14	[245,0]	[239,18]	19.485	TIP	BRANCH
15	[239,18]	[228,24]	12.485	BRANCH	BRANCH
16	[80,1]	[100,20]	29.213	TIP	BRANCH
17	[100,20]	[100,32]	11.0	BRANCH	BRANCH
18	[105,7]	[100,20]	13.657	TIP	BRANCH
19	[318,8]	[315,22]	14.243	TIP	BRANCH
20	[313,23]	[315,22]	1.0	TIP	BRANCH
21	[326,32]	[315,22]	15.485	TIP	BRANCH
22	[174,17]	[190,24]	19.314	TIP	BRANCH
23	[190,24]	[204,48]	29.385	BRANCH	BRANCH
24	[207,0]	[190,24]	29.627	TIP	BRANCH
25	[228,24]	[204,48]	35.284	BRANCH	BRANCH
26	[217,7]	[228,24]	20.728	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[57,0]	[54,25]	25.243	TIP	BRANCH
28	[54,25]	[61,43]	24.485	BRANCH	BRANCH
29	[35,34]	[54,25]	22.314	TIP	BRANCH
30	[356,12]	[353,27]	15.243	TIP	BRANCH
31	[369,20]	[353,27]	17.899	TIP	BRANCH
32	[353,27]	[339,43]	20.971	BRANCH	BRANCH
33	[100,32]	[71,40]	31.314	BRANCH	BRANCH
34	[100,32]	[106,72]	41.485	BRANCH	BRANCH
35	[70,38]	[71,40]	1.0	TIP	BRANCH
36	[71,40]	[61,43]	10.657	BRANCH	BRANCH
37	[326,45]	[339,43]	12.828	TIP	BRANCH
38	[339,43]	[335,65]	22.657	BRANCH	BRANCH
39	[171,44]	[172,47]	2.414	TIP	BRANCH
40	[172,47]	[201,51]	31.314	BRANCH	BRANCH
41	[172,47]	[187,64]	25.142	BRANCH	BRANCH
42	[204,48]	[204,51]	2.0	BRANCH	BRANCH
43	[201,51]	[204,51]	2.0	BRANCH	BRANCH
44	[201,51]	[201,64]	12.828	BRANCH	BRANCH
45	[213,59]	[204,51]	10.899	TIP	BRANCH
46	[341,65]	[366,61]	25.657	TIP	BRANCH
47	[366,61]	[366,64]	2.0	BRANCH	BRANCH
48	[371,57]	[366,61]	5.828	TIP	BRANCH
49	[187,64]	[201,64]	13.0	BRANCH	BRANCH
50	[183,83]	[187,64]	19.657	TIP	BRANCH
51	[201,64]	[205,86]	22.243	BRANCH	BRANCH
52	[376,65]	[366,64]	9.414	TIP	BRANCH
53	[363,66]	[366,64]	2.828	TIP	BRANCH
54	[307,65]	[333,66]	26.243	BRANCH	BRANCH
55	[308,72]	[307,65]	6.414	TIP	BRANCH
56	[304,56]	[307,65]	8.828	TIP	BRANCH
57	[335,65]	[333,66]	1.414	BRANCH	BRANCH
58	[337,66]	[335,65]	1.0	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[333,66]	[319,91]	29.385	BRANCH	BRANCH
60	[266,60]	[267,67]	6.414	TIP	BRANCH
61	[249,35]	[267,67]	50.971	TIP	BRANCH
62	[267,67]	[278,88]	24.142	BRANCH	BRANCH
63	[1,65]	[22,72]	28.314	TIP	BRANCH
64	[62,62]	[22,72]	44.799	TIP	BRANCH
65	[12,83]	[22,72]	14.314	TIP	BRANCH
66	[106,72]	[108,72]	1.0	BRANCH	BRANCH
67	[106,72]	[82,87]	29.799	BRANCH	BRANCH
68	[124,94]	[108,72]	31.142	TIP	BRANCH
69	[108,72]	[142,74]	35.899	BRANCH	BRANCH
70	[154,103]	[142,74]	33.799	TIP	BRANCH
71	[139,32]	[142,74]	50.941	TIP	BRANCH
72	[244,62]	[241,75]	13.243	TIP	BRANCH
73	[232,74]	[241,75]	8.414	TIP	BRANCH
74	[247,83]	[241,75]	9.485	TIP	BRANCH
75	[68,81]	[68,84]	2.0	BRANCH	BRANCH
76	[80,60]	[68,81]	24.556	TIP	BRANCH
77	[68,84]	[82,87]	14.243	BRANCH	BRANCH
78	[68,84]	[48,104]	28.627	BRANCH	BRANCH
79	[182,101]	[205,86]	30.556	TIP	BRANCH
80	[205,86]	[209,92]	6.243	BRANCH	BRANCH
81	[82,87]	[92,118]	35.556	BRANCH	BRANCH
82	[278,88]	[295,100]	22.728	BRANCH	BRANCH
83	[278,88]	[246,120]	48.527	BRANCH	BRANCH
84	[319,91]	[295,100]	26.728	BRANCH	BRANCH
85	[319,91]	[328,112]	23.314	BRANCH	BRANCH
86	[213,92]	[209,92]	3.0	TIP	BRANCH
87	[209,92]	[211,132]	40.657	BRANCH	BRANCH
88	[16,94]	[17,97]	2.414	TIP	BRANCH
89	[17,97]	[48,104]	33.728	BRANCH	BRANCH
90	[17,104]	[17,97]	6.414	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[295,100]	[296,109]	8.414	BRANCH	BRANCH
92	[48,104]	[69,110]	24.485	BRANCH	BRANCH
93	[307,112]	[296,109]	11.243	TIP	BRANCH
94	[298,145]	[296,109]	39.556	TIP	BRANCH
95	[74,102]	[69,110]	8.657	TIP	BRANCH
96	[69,110]	[92,118]	24.899	BRANCH	BRANCH
97	[316,107]	[328,112]	13.657	TIP	BRANCH
98	[92,118]	[92,121]	2.0	BRANCH	BRANCH
99	[246,120]	[249,128]	8.828	BRANCH	BRANCH
100	[246,120]	[242,128]	9.243	BRANCH	BRANCH
101	[115,151]	[92,121]	40.87	TIP	BRANCH
102	[86,172]	[92,121]	64.042	TIP	BRANCH
103	[240,128]	[242,128]	1.0	TIP	BRANCH
104	[242,128]	[249,128]	6.414	BRANCH	BRANCH
105	[269,118]	[249,128]	23.142	TIP	BRANCH
106	[19,114]	[19,131]	16.828	TIP	BRANCH
107	[0,112]	[19,131]	28.213	TIP	BRANCH
108	[19,131]	[19,134]	2.0	BRANCH	BRANCH
109	[211,132]	[231,153]	30.042	BRANCH	BRANCH
110	[211,132]	[207,136]	4.828	BRANCH	BRANCH
111	[36,159]	[19,134]	33.971	TIP	BRANCH
112	[6,153]	[19,134]	27.385	TIP	BRANCH
113	[205,137]	[207,136]	1.414	TIP	BRANCH
114	[207,136]	[203,183]	47.657	BRANCH	BRANCH
115	[257,178]	[231,153]	39.456	TIP	BRANCH
116	[237,171]	[231,173]	5.828	TIP	BRANCH
117	[230,179]	[231,173]	5.414	TIP	BRANCH
118	[230,171]	[231,173]	1.0	TIP	BRANCH
119	[458,183]	[458,181]	1.0	TIP	BRANCH
120	[431,141]	[458,181]	65.627	TIP	BRANCH
121	[216,196]	[203,183]	19.728	TIP	BRANCH
122	[193,193]	[203,183]	14.899	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[41,187]	[40,189]	1.414	TIP	BRANCH
124	[43,199]	[40,189]	10.828	TIP	BRANCH
125	[37,191]	[40,189]	2.828	TIP	BRANCH
126	[97,210]	[71,197]	45.556	TIP	BRANCH
127	[79,201]	[71,197]	9.243	TIP	BRANCH
128	[67,202]	[71,197]	5.243	TIP	BRANCH
129	[186,219]	[190,226]	7.657	TIP	BRANCH
130	[197,226]	[190,226]	6.0	TIP	BRANCH
131	[189,230]	[190,226]	3.414	TIP	BRANCH
132	[38,226]	[31,232]	10.243	TIP	BRANCH
133	[31,232]	[10,263]	39.284	BRANCH	BRANCH
134	[33,233]	[31,232]	1.0	TIP	BRANCH
135	[8,263]	[10,263]	1.0	BRANCH	BRANCH
136	[1,260]	[8,263]	7.657	TIP	BRANCH
137	[0,267]	[8,263]	8.243	TIP	BRANCH
138	[1,309]	[10,263]	51.799	TIP	BRANCH
139	[432,255]	[436,266]	11.657	TIP	BRANCH
140	[443,267]	[436,266]	6.414	TIP	BRANCH
141	[435,272]	[436,266]	5.414	TIP	BRANCH
142	[41,272]	[40,278]	5.414	TIP	BRANCH
143	[54,280]	[40,278]	13.828	TIP	BRANCH
144	[40,278]	[22,299]	27.627	BRANCH	BRANCH
145	[0,331]	[22,299]	44.213	TIP	BRANCH
146	[21,295]	[22,299]	3.0	TIP	BRANCH
147	[268,302]	[267,306]	3.414	BRANCH	BRANCH
148	[249,292]	[268,302]	22.314	TIP	BRANCH
149	[288,297]	[268,302]	20.657	TIP	BRANCH
150	[210,356]	[267,306]	83.154	TIP	BRANCH
151	[267,306]	[275,355]	52.314	BRANCH	BRANCH
152	[506,320]	[489,331]	21.142	TIP	BRANCH
153	[470,326]	[489,331]	20.071	TIP	BRANCH
154	[492,337]	[489,331]	6.243	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[275,355]	[285,365]	14.314	BRANCH	BRANCH
156	[272,356]	[275,355]	2.414	TIP	BRANCH
157	[290,364]	[285,365]	4.414	TIP	BRANCH
158	[283,367]	[285,365]	2.414	TIP	BRANCH
159	[115,397]	[125,399]	9.828	TIP	BRANCH
160	[138,415]	[125,399]	20.385	TIP	BRANCH
161	[129,398]	[125,399]	3.0	TIP	BRANCH
162	[36,416]	[37,422]	5.414	TIP	BRANCH
163	[37,422]	[48,425]	11.243	BRANCH	BRANCH
164	[1,441]	[37,422]	44.042	TIP	BRANCH
165	[57,418]	[48,425]	11.485	TIP	BRANCH
166	[36,440]	[48,425]	19.728	TIP	BRANCH
167	[57,457]	[55,460]	2.414	TIP	BRANCH
168	[39,475]	[55,460]	22.556	TIP	BRANCH
169	[61,461]	[55,460]	5.828	TIP	BRANCH
170	[67,473]	[73,472]	5.414	TIP	BRANCH
171	[75,471]	[73,472]	1.0	TIP	BRANCH
172	[73,472]	[83,479]	12.071	BRANCH	BRANCH
173	[93,471]	[83,479]	12.314	TIP	BRANCH
174	[126,511]	[83,479]	62.284	TIP	BRANCH
175	[31,494]	[38,501]	13.828	TIP	BRANCH
176	[37,503]	[38,501]	1.414	TIP	BRANCH
177	[48,510]	[38,501]	22.657	TIP	BRANCH
178	[107,0]	[159,39]	73.669	TIP	TIP
179	[467,11]	[471,24]	14.657	TIP	TIP
180	[400,29]	[385,41]	19.971	TIP	TIP
181	[483,53]	[485,64]	11.828	TIP	TIP
182	[416,55]	[435,55]	20.657	TIP	TIP
183	[460,55]	[456,74]	23.728	TIP	TIP
184	[414,57]	[422,62]	15.243	TIP	TIP
185	[471,105]	[471,120]	15.828	TIP	TIP
186	[181,113]	[164,129]	24.799	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[142,117]	[154,153]	41.799	TIP	TIP
188	[46,124]	[56,139]	19.728	TIP	TIP
189	[393,125]	[409,130]	18.071	TIP	TIP
190	[112,129]	[125,129]	13.828	TIP	TIP
191	[382,142]	[388,143]	6.414	TIP	TIP
192	[65,151]	[56,156]	16.728	TIP	TIP
193	[175,169]	[184,191]	42.799	TIP	TIP
194	[38,170]	[39,181]	12.243	TIP	TIP
195	[120,183]	[105,198]	24.142	TIP	TIP
196	[0,184]	[8,189]	10.657	TIP	TIP
197	[15,190]	[14,195]	5.414	TIP	TIP
198	[138,192]	[138,209]	19.485	TIP	TIP
199	[7,208]	[0,225]	20.485	TIP	TIP
200	[62,209]	[73,227]	28.657	TIP	TIP
201	[271,228]	[261,239]	15.728	TIP	TIP
202	[479,228]	[485,233]	8.657	TIP	TIP
203	[151,236]	[161,250]	20.728	TIP	TIP
204	[51,249]	[25,283]	48.87	TIP	TIP
205	[90,254]	[92,261]	7.828	TIP	TIP
206	[121,265]	[128,266]	8.243	TIP	TIP
207	[388,283]	[383,290]	12.243	TIP	TIP
208	[146,284]	[113,294]	37.142	TIP	TIP
209	[490,284]	[511,289]	23.071	TIP	TIP
210	[439,302]	[441,375]	85.87	TIP	TIP
211	[209,319]	[191,323]	19.657	TIP	TIP
212	[51,321]	[63,327]	15.071	TIP	TIP
213	[496,343]	[511,373]	36.213	TIP	TIP
214	[36,347]	[50,352]	16.071	TIP	TIP
215	[22,348]	[12,368]	26.142	TIP	TIP
216	[148,353]	[140,364]	14.314	TIP	TIP
217	[268,361]	[268,369]	9.657	TIP	TIP
218	[110,363]	[125,366]	17.657	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[378,366]	[383,373]	9.657	TIP	TIP
220	[66,371]	[63,377]	7.243	TIP	TIP
221	[296,376]	[297,390]	14.414	TIP	TIP
222	[334,388]	[344,394]	12.485	TIP	TIP
223	[180,397]	[175,416]	21.071	TIP	TIP
224	[81,399]	[119,442]	63.426	TIP	TIP
225	[412,405]	[423,416]	17.314	TIP	TIP
226	[140,418]	[148,437]	23.142	TIP	TIP
227	[384,446]	[350,462]	41.213	TIP	TIP
228	[198,456]	[217,482]	39.87	TIP	TIP
229	[15,475]	[17,483]	8.828	TIP	TIP
230	[282,500]	[306,510]	29.899	TIP	TIP